

Intesar Alam

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phase 2, Islamabad, Pakistan

EDUCATION

Ghulam Ishaq Khan Institute of Engineering Sciences & Technology, Topi	2022 - 2026
- Bachelor of Science – Mechanical Engineering	
Islamia College, Peshawar	2019 - 2021
- FSc	
Peshawar Model School Boys-1, Peshawar	2017 -2019
- Matriculation	

PROFESSIONAL EXPERIENCE

IT MART- Peshawar	Jan 2020 – Present
<i>Supply Chain Manager (Part-Time)</i>	
- Led demand forecasting and inventory planning for a retail store with ~2,000+ SKUs, reducing stockouts by 9.5% and cutting excess inventory by 6% through data-driven reorder point optimization and seasonal sales analysis. - Managed end-to-end logistics and supplier relationships with 21+ vendors, negotiated pricing and delivery terms to lower procurement costs by 5.3%.	
Tarbela Power Station (WAPDA)	Jul 2025 – Sep 2025
<i>Mechanical Intern</i>	
- Studied hydropower plant operations including turbines, governors, pumps, and auxiliary systems. - Gained exposure to mechanical drawings, preventive maintenance, inspection routines, and industrial safety practices.	
Oil & Gas Development Company Limited (OGDCL) – Tough Gas Field	Jun 2025 – Jul 2025
<i>Mechanical Intern</i>	
- Gained on-site exposure to gas production operations, rotating equipment, and maintenance workflows. - Observed a full well intervention operation, gaining practical insight into the step-by-step procedure, equipment setup, safety protocols, and coordination required during live gas field activities.	

LEADERSHIP & STUDENT EXPERIENCE

Team Invictus – Student-run technical team competing in the AIAA DBFC	Dec 2022 – May 2024
<i>Sub-Head of Design team</i>	
- Contributed to aircraft design and analysis using SolidWorks and XFLR5. - Engaged in fabrication sessions of the aircraft itself.	
<i>Sub-Head of Documentation team</i>	
Part of documentation for proposal submission resulting in achieving the 5th highest proposal score worldwide	
IEEE GIKI Student Branch – Student branch of IEEE	Oct 2022 – Sep 2024
<i>Sub-Head of liaison team</i>	
As a member of the liaison team for the esteemed National Electronic Olumpia’23 & National Electronic Olumpia’24(Sub-Head) played a vital role in facilitating seamless communication and collaboration among diverse stakeholders	
<i>Sponsorship Member</i>	
Facilitated in marketing and sponsorship	
AIAA GIKI Student Branch – Student branch AIAA	Feb 2023 – Dec 2023
<i>Sub-Head of NDBFC’17</i>	
Successfully managed the execution of the National Design and Build Competition, a prestigious all-Pakistan event where university teams design, build, and compete with RC planes. Oversaw event logistics, facilitated team coordination, and ensured seamless operations, contributing to a highly engaging and well-received competition.	
Event Organizing Society – Student based society	Sep 2018 – Apr 2019
<i>Vice President</i>	

- Played a key role in planning and executing major events, including the Annual Day and Sports Gala. Responsibilities included coordinating with teams, managing logistics, and ensuring the successful delivery of high-profile events.

PUBLICATIONS

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- **Co-author — *Designing a Sustainable Organic Rankine Cycle for Remote Geothermal Heat Sources in Pakistan*** (Mater. Proc., MDPI; presented at MTME 2025). Designed and analysed a single-stage ORC using low-GWP R600a, performing parametric studies on temperature, pressure, and mass flow to maximize net power (≈ 11.7 kW) and cycle efficiency (peak $\approx 8.28\%$), while minimizing component exergy destruction.
 - **Co-author — *Bio-Inspired Design and CFD Performance Analysis of a Hydrofoil Turbine for Diverse Marine Applications*** (manuscript under review; presented at MTME 2026). Designed and optimized a 0.17 m chord hydrofoil using AI-inspired geometry and ANSYS CFX, performing parametric studies on inlet velocity (0.94–10.6 m/s), angle of attack (0° – 10°), and Reynolds number ($\approx 1.6 \times 10^5$ – 1.8×10^6) to maximize aerodynamic performance, achieving peak lift-to-drag ratio ≈ 13.1 and up to $\approx 61\%$ improvement over baseline design while ensuring mesh-independent results.
 - **Co-author — *Thermal Image-Based Fault Detection in Induction Motors Under Limited Data Conditions (Awarded Best Paper Award for MTME 2026)*** (manuscript under review; presented at MTME 2026). Developed a transfer learning-based fault classification framework using pretrained SqueezeNet CNN on a limited dataset (120 thermal images), performing stratified 5-fold cross-validation to classify four motor conditions, achieving mean accuracy $\approx 97.1\%$ ($\pm 2.3\%$) and macro F1-score ≈ 0.97 , outperforming SVM baseline ($\approx 82.5\%$) while ensuring robust evaluation against data leakage.
 - **• Co-author, *Unsupervised Bearing Fault Detection Using Time-Domain Features and Mahalanobis Distance Scoring*** (manuscript under review; presented at MTME 2026). Developed an unsupervised bearing condition-monitoring framework using RMS, standard deviation, and kurtosis features with multivariate Gaussian modeling and Mahalanobis distance scoring for anomaly detection from healthy data only. Evaluated on the CWRU bearing dataset, the method achieved a **macro AUC of 0.985** and **macro F1-score of 0.978** across three inner-race fault severities, with a lightweight design suitable for real-time monitoring.
 - **• Co-author, *Physics-Informed Neural Network for State of Health Forecasting of Electric Vehicle Batteries*** (manuscript under review; presented at IBCAST 2026). Developed a lightweight physics-informed neural network for one-step-ahead EV battery SoH forecasting using NASA battery data, incorporating monotonicity and endpoint-anchor constraints to improve physical consistency and prediction accuracy. The proposed model outperformed MLP, LSTM, and TCN baselines, achieving **MAE = 1.43%**, **RMSE = 1.89%**, and **$R^2 = 0.9931$** .

ACADEMIC PROJECTS

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- **Hybrid Multi-Stage Turbine for Vehicle-Induced Wind Energy (Final Year Project – Ongoing)**
Designed and simulated a hybrid turbine system to harvest wind energy generated by moving vehicles. Performed aerodynamic design, structural modeling, and performance evaluation using CREO, SolidWorks, and ANSYS, with emphasis on efficiency, scalability, and infrastructure-level deployment.
 - **Thermodynamic Design and Exergy Analysis of a 150 kW Brayton–ORC Power Plant**
Designed a combined Brayton–Organic Rankine Cycle power plant and conducted component-level energy and exergy analysis. Identified major irreversibilities in the combustor and HRSG, performed parametric studies on ORC boiler pressure, and improved net output and overall efficiency.
 - **Shell-and-Tube Heat Exchanger CFD Analysis (COMSOL Multiphysics)**
Performed CFD analysis of turbulent shell-side flow using RANS k- ϵ modeling in COMSOL Multiphysics 6.3. Evaluated four design variants with fin geometries and tube arrangements, quantified pressure drop and outlet temperature trade-offs, and recommended optimal configurations for balanced thermal performance.
 - **ANN-Based Predictive Maintenance Classification**
Built an ANN model to predict machine failure from operational sensor data. Performed feature analysis, model training, and validation using performance metrics and confusion analysis to enable data-driven maintenance decisions.
 - **Disaster-Response Robot for Emergency Supply Delivery**
Designed and developed a mobile disaster-response robot using additive manufacturing to deliver first-aid supplies in inaccessible areas. Conducted structural design, CAD modeling, and load analysis in SolidWorks to ensure operational reliability in emergency scenarios.
 - **Experimental Analysis of Bearing Faults Using Vibration and Acoustics**
Investigated rolling-element bearing defects through experimental measurement of vibration, sound, and electrical

signals. Analyzed frequency-domain responses across multiple operating speeds to support condition-monitoring methodologies.

- **Industrial Conveyor System Design and Power Analysis**

Carried out mechanical design of a belt-driven conveyor for material handling applications. Calculated power, torque, and component sizing, developed CAD assemblies, and assessed manufacturability and cost considerations.

ADDITIONAL ACADEMIC & LAB PROJECTS

- **Induction Motor Flux Control via Winding Configuration Switching**

Simulated induction motor behaviour using electromagnetic modelling to study flux-weakening through alternative winding connections. Compared performance metrics such as torque, voltage, and magnetic flux under different switching strategies.

- **Statically Indeterminate Shaft under Torsional Loading (Analytical + FEA)**

Designed and analyzed a multi-segment statically indeterminate shaft under torsional loads using analytical methods and Finite Element Analysis in ANSYS. Evaluated stress distribution, angle of twist, and factor of safety to validate design assumptions.

- **Lightweight Structural Bracket Optimization Using FEA**

Applied topology optimization techniques to redesign a metallic bracket for reduced mass while maintaining structural integrity. Verified optimized designs using finite element stress and deformation analysis.

- **PID-Based Temperature Control Using Arduino**

Designed and implemented a closed-loop PID temperature control system using Arduino, thermocouple feedback, and PWM-driven heating. Modeled system dynamics in Simulink, tuned controller gains, and validated performance through hardware testing under sensor delay and thermal lag.

- **Spring-Mass Accelerometer Modeling in MATLAB**

Designed a mechanical accelerometer model based on spring-mass dynamics for low-g acceleration measurement. Simulated system response and studied sensitivity and parameter effects using MATLAB.

- **Effect of Electrode Size on Droplet Velocity in a DC-Actuated EWOD Microfluidics Chip**

Analysed droplet transport in a DC-actuated EWOD system by comparing small and large electrode geometries. Measured step times via video tracking and evaluated normalized velocity to study the effect of electrode size and electric field distribution on droplet motion.

- **Surface Tension Measurement Through Capillary Flow Testing**

Performed laboratory experiments to estimate liquid surface tension using capillary wicking in porous media. Processed experimental data using image analysis and curve fitting techniques.

- **Design and Experimental Validation of Active Low-Pass and High-Pass Filters**

Designed first-order active LPF and HPF circuits using op-amps, derived cutoff frequency analytically, and validated frequency response through Simulink modeling and hardware experiments.

- **Autonomous Line-Tracking Robot Using Embedded Control**

Built a microcontroller-based mobile robot capable of following predefined paths using infrared sensing. Integrated mechanical design, electronic circuitry, and control logic for real-time navigation.

CERTIFICATIONS

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| - MATLAB Onramp (MathWorks) | - The Bits and Bytes of Computer Networking (Google) |
| - Simulink Onramp (MathWorks) | - Technical Support Fundamentals (Google) |
| - Calculations with Vectors and Matrices (MathWorks) | - Introduction to ANSYS Fluent (ANSYS) |

TECHNICAL SKILLS

- **Engineering Software & Analysis Tools:** AutoCAD, ANSYS CFX and Maxwell, COMSOL Multiphysics, SolidWorks, MATLAB, Simulink, EES, Origin, Proteus, Python (NumPy, Pandas, Matplotlib), YOLOv8, TensorFlow/Keras, computer-aided design, computational fluid dynamics, finite element analysis
- **Control, Programming & Data Handling:** PID-based control systems, Arduino–MATLAB data acquisition, embedded C/C++ development, machine learning techniques (classification, regression, convolutional neural networks), frequency-domain and FFT-based signal analysis

- **Mechanical Systems & Mechatronics:** Mechanical mechanism synthesis, shaft and drivetrain design, conveyor system development, topology-driven structural optimization, kinematic analysis, electromechanical system integration, BLDC motor actuation, pneumatic and rotary system concepts
- **Manufacturing & Product Realization:** Additive manufacturing, workshop fabrication practices, CNC machining fundamentals, evaluation of cost, manufacturability, and production feasibility
- **Laboratory, Field & Industrial Exposure:** Oscilloscopes, vibration and acoustic sensing equipment, current measurement techniques, wind-tunnel testing exposure, industrial field exposure at WAPDA hydropower facilities and Oil & Gas Development Company Limited (OGDCL) gas field operations, including equipment observation, maintenance practices, and HSE-compliant workflows
- **Core Engineering Knowledge:** Solid and rigid-body mechanics, machine element design, stress and vibration analysis, robotics and automation, electrical machines and drives, thermodynamics, fluid and thermal sciences, turbomachinery, manufacturing processes, system dynamics and control, thermo-fluid systems, instrumentation and digital twin concepts, numerical methods, linear algebra, differential equations, probability and statistics, engineering economics, project execution, occupational safety, and technical communication

ADDITIONAL INFORMATION

- Volunteer work: **United Nations** Association of Pakistan (Youth Member)
- Languages: English (written and spoken proficiency), Urdu (written and spoken proficiency), and Pashto (native)